VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024 — Surpassing DAPO by 10+ Points

ByteDance Seed | Qwen2.5-32B Base | No SFT Data

arXiv:2504.05118

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO / DAPO)

- Compute advantage from final trajectory reward
- Assign same advantage uniformly to every token
- Works for simple tasks, but breaks down for multi-step reasoning
- Step-level errors receive no differentiated penalty

Value-Model-Based (VAPO)

- Token-level credit assignment via value estimates
- Precisely trace each action's impact on returns
- Lower variance per token vs Monte Carlo
- Better generalization and sample efficiency

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

1. Value Model Bias

- Long trajectories amplify bootstrapped value estimation errors.
- Bias compounds across thousands of tokens.
- Standard GAE struggles to stay stable.

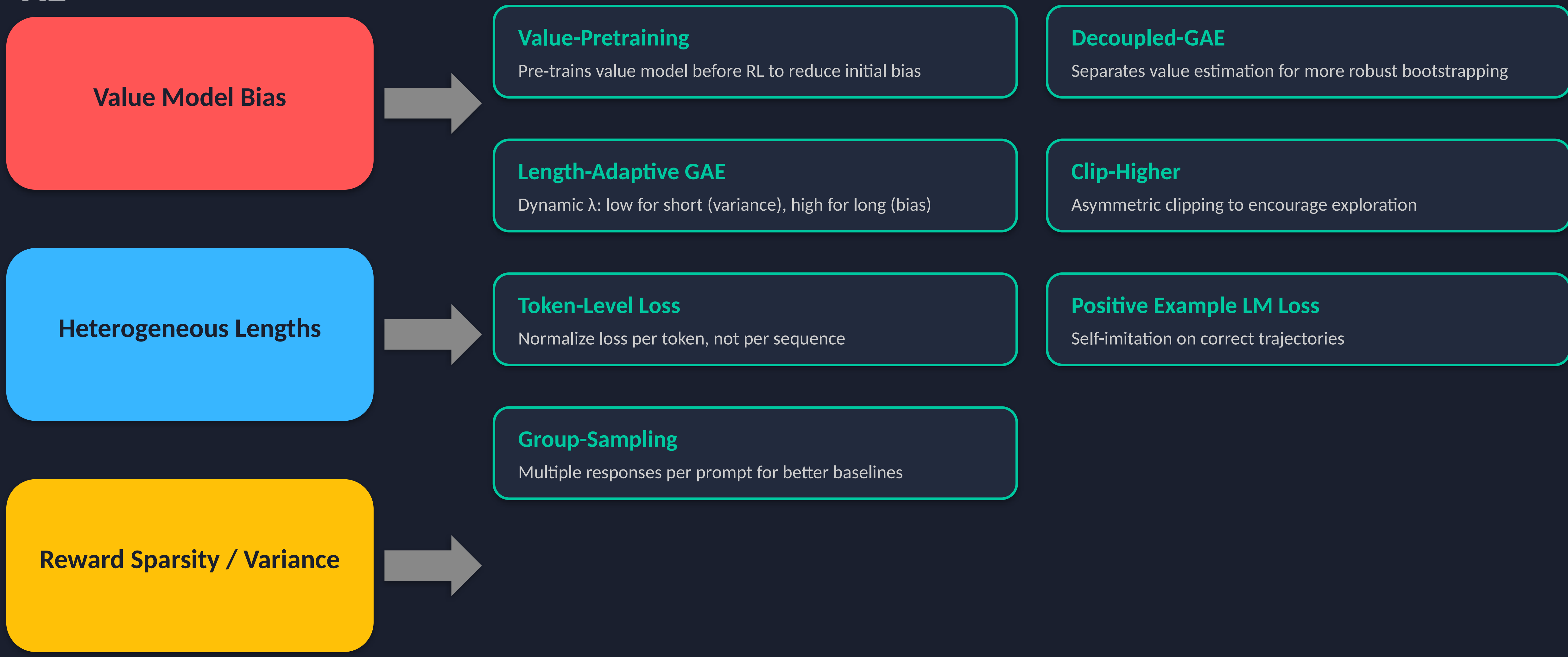
2. Heterogeneous Lengths

- Responses vary from hundreds to thousands of tokens.
- Fixed λ forces a single bias-variance tradeoff.
- Short and long sequences need different treatment.

3. Reward Sparsity

- Only the final <eos> token receives reward.
- Sparse signal makes credit assignment hard.
- Value model must generalize across rare rewards.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL



Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Core Idea

- GAE λ controls bias-variance tradeoff:
  - $\lambda \to 0$: high bias, low variance (rely on value model)
  - $\lambda \to 1$: low bias, high variance (rely on returns)
- Fixed λ cannot serve short and long responses simultaneously
- Length-Adaptive GAE sets λ as a function of sequence length
  - Short responses → lower λ (reduce variance)
  - Long responses → higher λ (reduce bias)

Short vs Long Response Behavior

Short Response

Low λ
↓ variance
Rely on value model

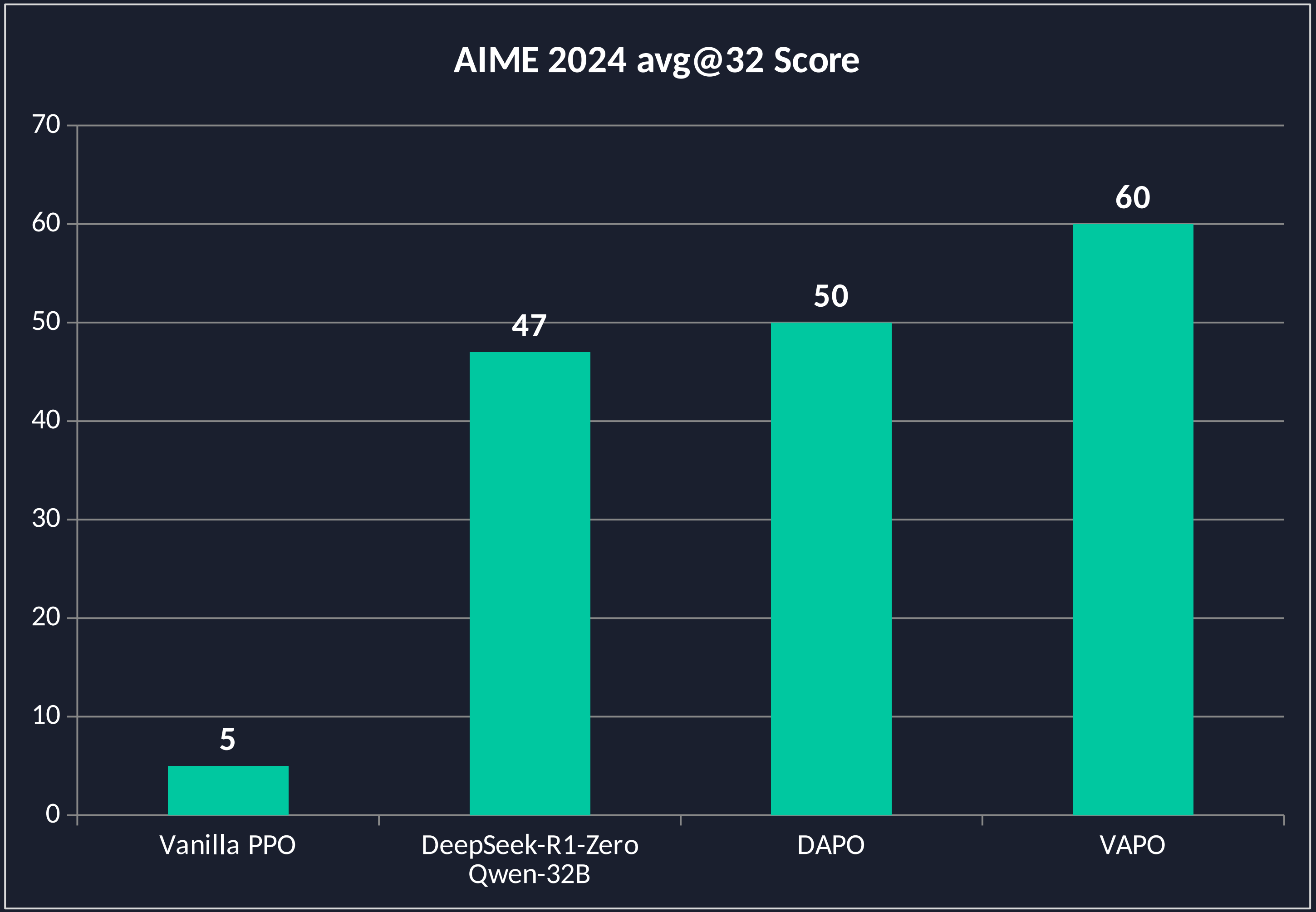
Long Response

High λ
↓ bias
Rely on returns

Impact: Removing Length-Adaptive GAE drops AIME score from 60 → 45

This is a novel contribution addressing a problem unique to long-CoT reasoning.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



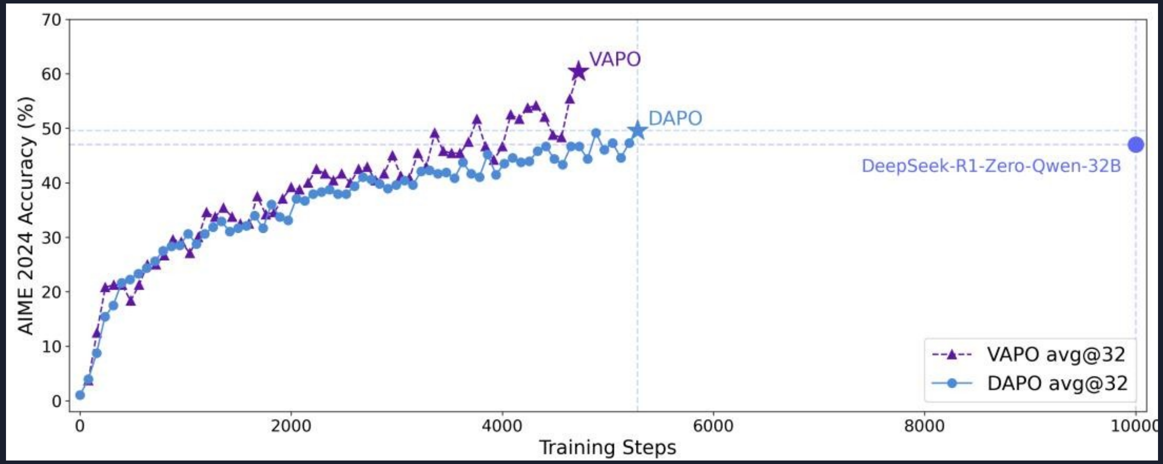
Key Results

- VAPO: 60.4 avg@32
- DAPO: 50 (+10.4 gap)
- DeepSeek-R1-Zero: 47
- Vanilla PPO: 5

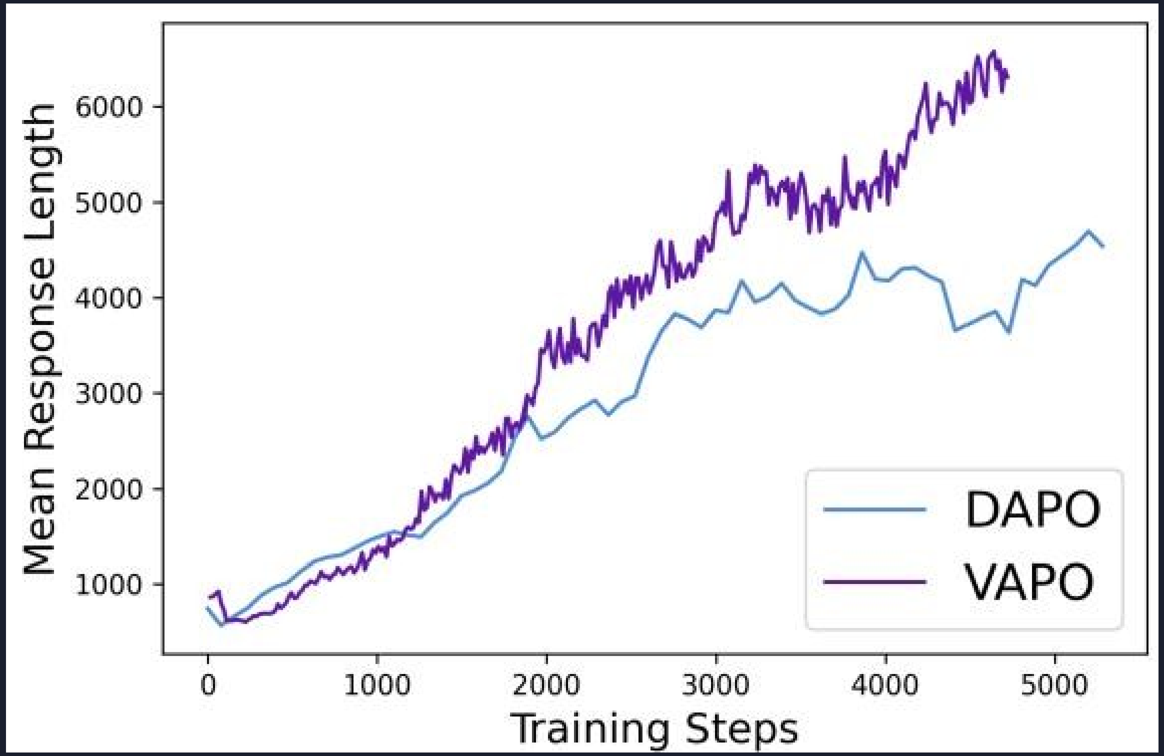
All methods use Qwen2.5-32B with no supervised fine-tuning.

VAPO reaches SOTA in ~5,000 steps with zero training crashes.

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO



Training accuracy: VAPO vs DAPO on AIME 2024



Mean response length over training

VAPO hits 60% at step ~4,500 | DAPO plateaus near 50% at step ~5,500 | VAPO generates ~6,000 tokens vs DAPO ~4,500

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME 2024 avg@32	Δ from VAPO
VAPO (full)	60	—
VAPO w/o Positive Example LM Loss	54	−6
VAPO w/o Token-level Loss	53	−7
VAPO w/o Group-Sampling	55	−5
VAPO w/o Clip-Higher	46	−14
VAPO w/o Length-Adaptive GAE	**45**	**−15**
VAPO w/o Decoupled-GAE	**33**	**−27**
VAPO w/o Value-Pretraining	**11**	**−49**

Critical: Removing Value-Pretraining drops score from 60 → 11. It is the single most impactful component.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1. Value-model-based RL can surpass value-free methods

When bias, heterogeneous lengths, and sparsity are systematically addressed.

2. Length-Adaptive GAE is critical for long-CoT reasoning

Novel dynamic λ adjustment contributes 15 points; fixed λ fails across variable lengths.

3. Value-Pretraining is the single most important component

Contributes 49 points. Without it, value-model-based RL barely works on long-CoT tasks.

4. VAPO is highly stable and efficient

Reaches SOTA in ~5,000 steps with zero training crashes across multiple independent runs.

5. Implications for future reasoning models

Token-level credit assignment and adaptive GAE should be standard tools for long-horizon LLM RL.